SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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Technical Presentation

Architecture of Data Warehouse

CSA1606 - DATA WAREHOUSING AND DATA MINING

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INTRODUCTION

What is the Architecture of a Data Warehouse?

- Data Warehouse Architecture is the structure that defines how data is collected, processed, stored, and accessed.
- It organizes data from different sources into a central repository using the ETL process.
- It helps users analyse historical data and generate reports for decision-making.

Importance

- Stores historical data.
- Supports business decisions.
- Improves reporting and analysis.
- Helps identify trends and patterns.

Core Concepts

Three-Tier Architecture

- A Data Warehouse uses a **three-tier architecture** to collect, store, and analyze data efficiently.

1. Bottom Tier

- Collects data from different sources.
- Uses the ETL process (Extract, Transform, Load).
- Stores data in the Data Warehouse.

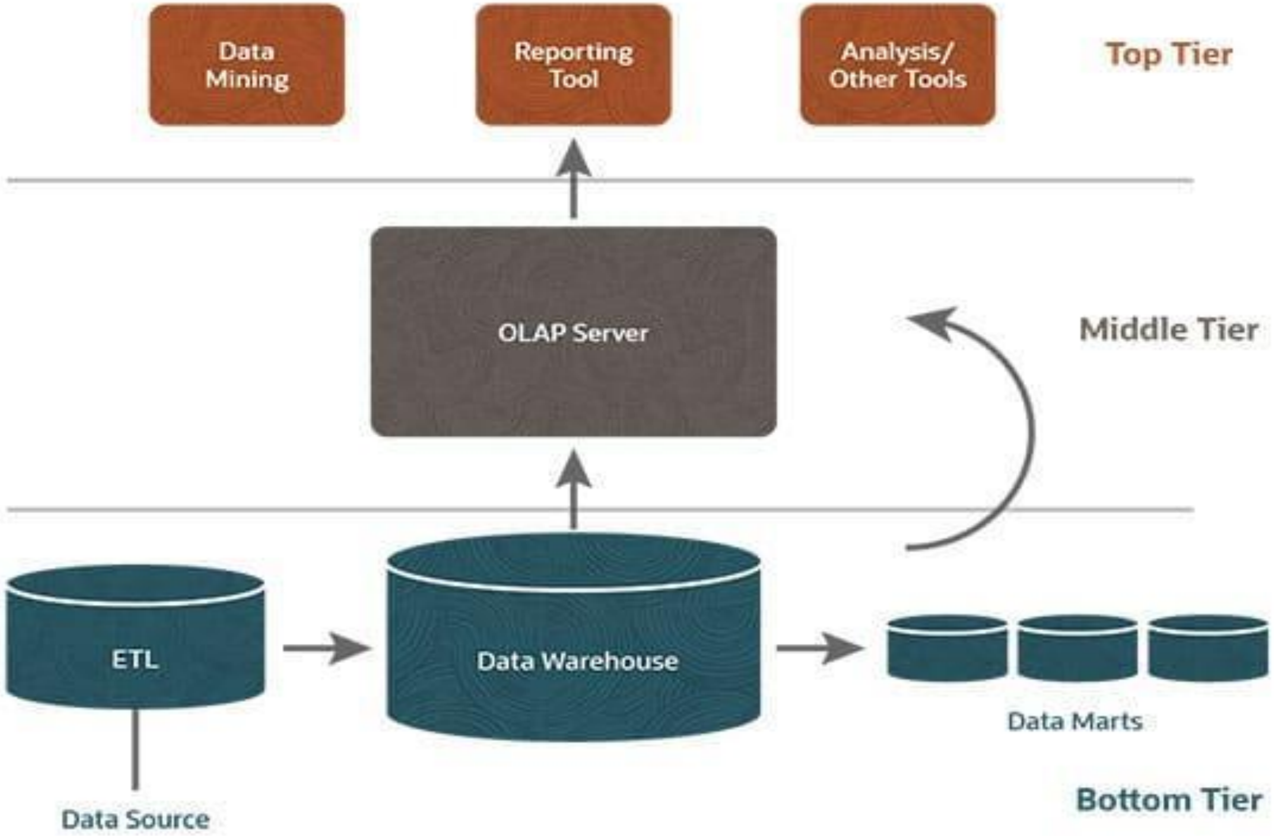
2. Middle Tier

- Contains the OLAP Server.
- Organizes and processes data for analysis.

3. Top Tier

- Used by users to access reports and dashboards.
- Includes Business Intelligence (BI) tools for data analysis.

Diagram



Data Mining
Reporting Tool
Analysis/ Other Tools
Top Tier
OLAP Server
Middle Tier
ETL
Data Warehouse
Data Marts
Data Source
Bottom Tier

ETL PROCESS

ETL (Extract, Transform, Load)

ETL is the process of collecting data from different sources, converting it into a useful format, and storing it in the Data Warehouse.

1. Extract

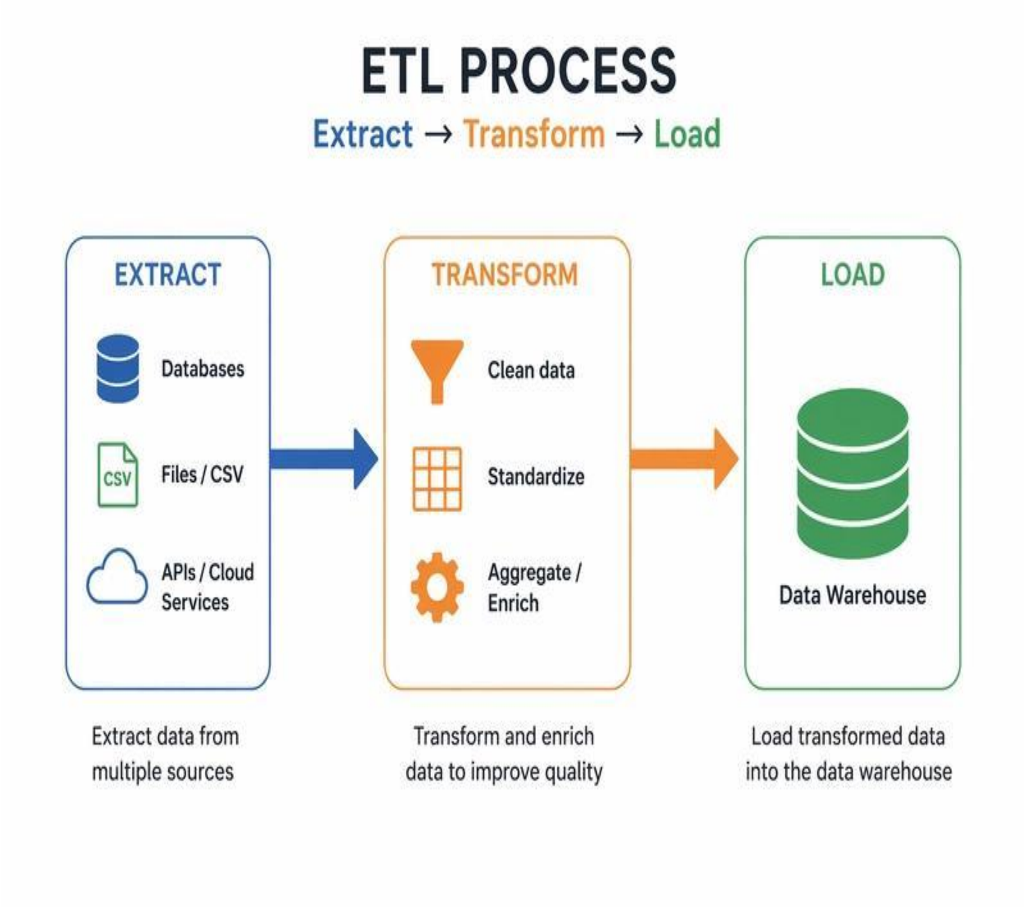
- Collects data from different sources.
- Sources include databases, Excel files, websites, and business applications.

2. Transform

- Cleans incorrect or duplicate data.
- Converts data into a common format.
- Improves data quality.

3. Load

- Loads the transformed data into the Data Warehouse.
- The data is then ready for analysis and reporting.



Data Mart and Metadata

Data Mart

- A small part of a Data Warehouse designed for a specific department.
- Stores department-specific data for quick analysis.
- Examples:
  - Sales Data Mart
  - Finance Data Mart
  - HR Data Mart

Metadata

- Metadata means **"data about data"**.
- Provides information about data source, format, and location.
- Helps users understand and manage stored data.

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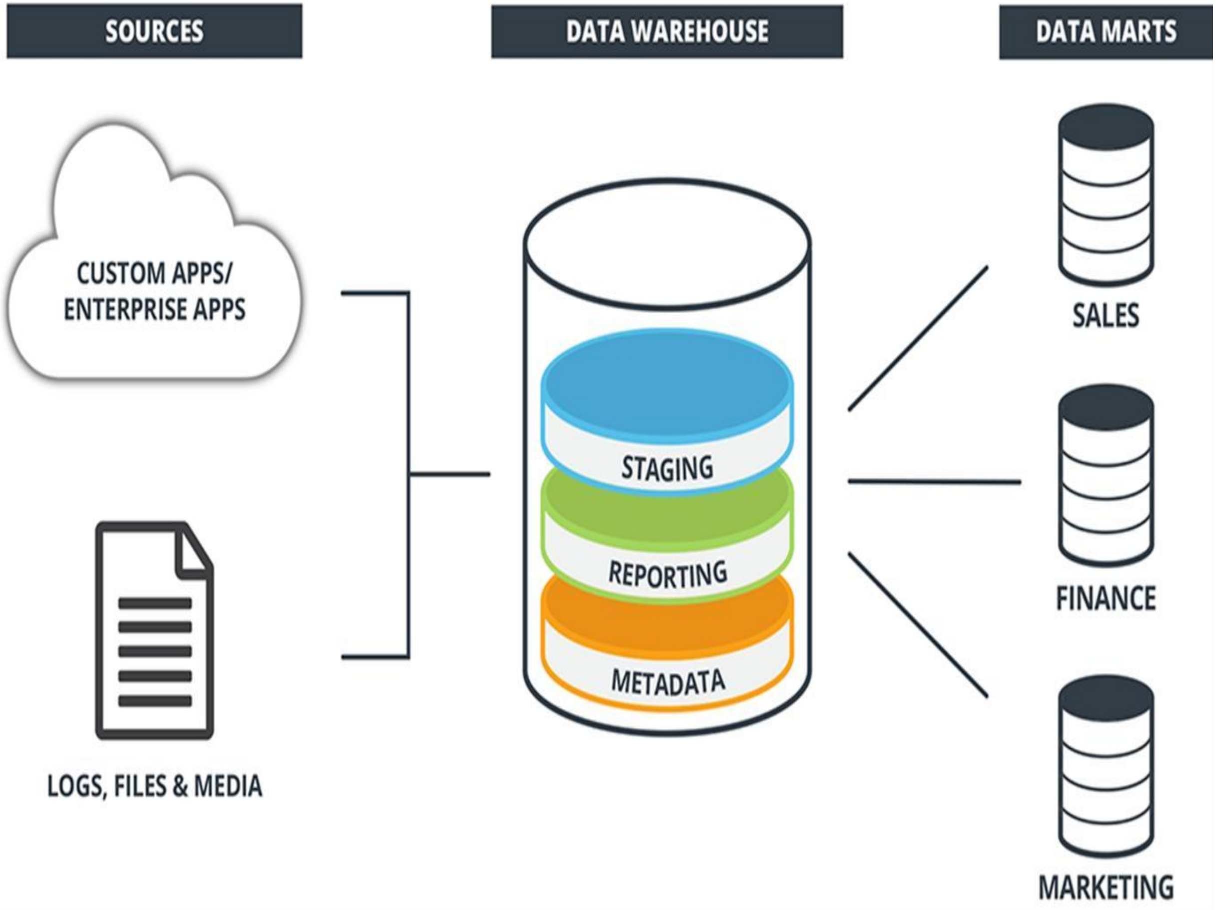
Metadata Example

Data: Customer Name, ID, Purchase Details

Metadata:

- Data source: Customer Database
- Data type: Text/Number
- Last updated: July 2026



Working Process of Data Warehouse

Step-by-Step Process

1.Data Collection

•Data is collected from different sources like databases, ERP, and CRM.

2.ETL Process

•Data is extracted, cleaned, and transformed into a proper format.

3.Data Storage

•Processed data is stored in the Data Warehouse.

4.Data Analysis

•OLAP tools analyze the stored data.

5.Report Generation

•Reports and dashboards are created for decision-making.

Real-World Applications

- A Data Warehouse is used in different industries to store, analyze, and manage large amounts of data for better decision-making.

1. Banking

- Analyzes customer transactions.
- Detects fraud activities.
- Helps in loan and risk analysis.

2. Healthcare

- Stores patient records.
- Analyzes disease patterns.
- Improves hospital management.

3. Retail

- Analyses sales data.
- Tracks customer behaviour.
- Helps in inventory management
- **4. Education**
- Analyses student performance.
- Maintains academic reports.

Tools Used in Data Warehouse

KNIME – Creates data workflows and performs data analysis.

Power BI – Creates dashboards and visual reports.

Tableau – Provides data visualization and analytics.

PostgreSQL/MySQL – Stores and manages data.

CONCLUSION

- A Data Warehouse helps organizations store and analyse large amounts of data from different sources.
- It improves decision-making by providing accurate reports and useful insights.
- It supports Business Intelligence (BI), analytics, and future technologies like AI and Machine Learning.
- Data Warehouses will continue to play an important role in managing and analysing data efficiently.

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Thank You